

Comprehensive Work Summary Report

Radar-Camera Fusion System for Maritime Small Object Detection

Prepared for: Reference Letter

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1 Executive Summary

This report provides a detailed chronological analysis of work completed on a maritime radar-camera fusion system project, spanning from March 2025 through October 2025. The project addresses the critical challenge of detecting small, low-radar-cross-section (RCS) objects at sea by combining radar and optical sensors with machine learning approaches.

Over an approximately seven-month period, the candidate has:

- Completed comprehensive literature review covering radar fundamentals, signal processing, and sensor fusion
- Designed an end-to-end sensor fusion architecture integrating radar and camera modalities
- Developed multi-gain radar data processing techniques
- Initiated CNN model development for radar-based object detection
- Established preliminary data collection pipeline at Lake Murray
- Explored numerous technical approaches including integer processing, entropy analysis, tracking algorithms, and neural network architectures
- Planned the A-MARV (Amplitude-only Maritime Radar Video) dataset initiative

The work demonstrates strong technical depth, systematic project planning, and iterative refinement. Some aspects remain in-progress, with challenges primarily stemming from the inherent complexity of multi-sensor calibration and the novelty of applying deep learning to maritime radar data.

2 Project Overview and Context

2.1 Problem Statement and Motivation

Marine radar systems struggle to reliably detect small, low-RCS targets (partially submerged swimmers, floating buoys, debris) against dynamic ocean backgrounds. These objects produce weak radar echoes easily masked by sea clutter. Simultaneously, camera-based deep learning approaches for maritime object detection face a severe data bottleneck—collecting and annotating small objects in maritime scenes is prohibitively expensive.

The project proposes to overcome these limitations through sensor fusion: leveraging radar’s all-weather detection capabilities to cue camera-based systems, while using

the camera's high-resolution detail to classify and validate radar detections. Additionally, radar detections can serve as automatic region proposals for image labeling, reducing manual annotation effort by potentially an order of magnitude.

2.2 Project Scope

The research encompasses multiple technical areas:

1. **Radar Signal Processing:** Gain control, CFAR thresholding, multi-gain data fusion
2. **Sensor Calibration:** Co-registration of radar and camera coordinate frames
3. **Machine Learning:** CNN development for radar object detection, possibly incorporating Vision Transformers
4. **Data Generation:** Building annotated datasets for training and evaluation
5. **Systems Integration:** End-to-end pipeline from raw sensor data to labeled outputs

3 Chronological Work Summary

3.1 Phase 1: Project Conception and Literature Review (March – May 2025)

3.1.1 Initial Project Idea Presentation (March 2, 2025)

The project was formally initiated with presentation of the core concept. The initial presentation established:

- **System setup:** Radar and camera positioned to observe marine objects at known ranges and bearings
- **Output modalities:** Radar provides polar coordinate detections; camera provides Cartesian image-space localization
- **Key insight:** By combining radar (range/bearing, all-weather) and camera (high-resolution classification), both sensor limitations can be overcome
- **Novelty:** Using radar not just for detection, but as an automatic labeling cue for camera datasets

Status: COMPLETED. The initial concept presentation was thorough and well-articulated, with clear diagrams distinguishing radar-only, camera-only, and fused detection paradigms. This foundational clarity enabled subsequent technical work.

3.1.2 Comprehensive Literature Review (May 2025)

Status: COMPLETED. A formal literature review was conducted, spanning:

- **Marine radar fundamentals:** Operating modes (pulsed vs. FMCW), frequency bands (X-band vs. S-band)
- **Detection challenges:** Sea clutter characteristics, low-RCS target physics, grazing angle effects
- **Signal processing techniques:** CFAR detection, Doppler filtering (MTI), pulse compression, high-resolution techniques
- **Computer vision methods:** Faster R-CNN, YOLO, other object detection architectures
- **Sensor fusion approaches:** Multi-sensor architectures in autonomous driving, maritime systems
- **Data scarcity problem:** Maritime dataset limitations, few-shot learning approaches

The review synthesized 131+ references into a coherent problem statement with well-motivated research objectives. This demonstrates systematic understanding of the problem space and positions the work within existing literature.

Observation: The literature review is extensive and well-structured, indicating thorough preliminary research. However, no novel literature contributions (e.g., identifying gaps in prior work) appear to have been generated at this stage.

3.2 Phase 2: Technical Foundation and Exploration (May – September 2025)

This phase involved both conceptual refinement and hands-on technical exploration, evidenced by monthly progress reports and frequent technical presentations.

3.2.1 Radar Signal Processing Fundamentals (May – June 2025)

Work on Gain Control and CFAR:

The candidate developed detailed understanding of radar gain and detection thresholding:

- **Manual gain tuning challenges:** Demonstrated how single fixed gain settings fail across varying clutter conditions

- **Automatic Gain Control (AGC) limitations:** Identified that AGC, while adaptive, indiscriminately suppresses small targets along with background
- **CFAR implementation:** Implemented and analyzed Constant False Alarm Rate detection on Lake Murray data
- **Multi-gain sampling methodology:** Conceptualized capturing radar data at 10 different gain levels (5%, 15%, 25%, ..., 95%) to span dynamic range

Status: COMPLETED (foundational understanding); PARTIALLY COMPLETED (implementation/validation).

Working implementation of CFAR detector on real radar data is demonstrated (Figure 7 in documents shows CA-CFAR output). However, large-scale validation across diverse sea states appears incomplete.

Presentations on Gain Timing and Comparison (June 7 and July 7, 2025):

Two presentations specifically addressed gain optimization:

- Gain timing analysis comparing detection performance across gain levels
- HDR-style gain comparison showing how combining multiple exposures enables target visibility across dynamic range

Status: PARTIALLY COMPLETED. Concepts were explored and presented, but quantitative performance metrics comparing gain strategies not clearly demonstrated in available documentation.

3.2.2 Radar Data Analysis and Processing (July – August 2025)

Radar Entropy Analysis (July 21, 2025):

Presentation on radar entropy as potential metric for sea clutter characterization.

Status: PARTIALLY COMPLETED. Entropy was explored as a potential feature, but integration into detection pipeline not clearly established in deliverables.

Integer Representation Work (July 2 and October 22, 2025):

Multiple presentations on integer processing of radar data, exploring efficient numerical representations.

Status: UNCLEAR COMPLETION. While presentations were delivered, practical impact on the main detection pipeline is not documented.

Radar Calibration and Simulation (July 30, 2025):

Work on sensor calibration and radar simulation was initiated.

Status: IN PROGRESS. Calibration is critical for sensor fusion but remains a challenging ongoing task. No final calibration results presented.

3.2.3 Machine Learning Infrastructure (August – September 2025)

Object Detection Exploration (August 14, 2025):

Initial exploration of detection architectures.

Status: PARTIALLY COMPLETED. Architectures reviewed but no training results on radar data presented.

Object Tracking and MAE Work (August 19, 2025):

Work on object tracking algorithms and Masked Autoencoder (MAE) approaches.

Status: IN PROGRESS. MAE preliminary results shown (Figure 15 in document shows MAE reconstruction with “~10 minutes on MacBook Air”), indicating early-stage experimentation. This represents novel work in applying self-supervised learning to radar.

GPU Sharding and Computational Infrastructure (September 23-25, 2025):

Presentations on GPU optimization and distributed training frameworks.

Status: PARTIALLY COMPLETED. Infrastructure explorations conducted, but unclear if integrated into production pipeline.

3.2.4 Monthly Progress Reports (May – September 2025)

Five monthly Phase 2 reports documented progress:

- **May 2025:** Project initiation, literature review finalization, conceptual framework
- **June 2025:** Gain control analysis, team coordination, initial data collection planning
- **July 2025:** Multiple technical explorations (entropy, integer, calibration, tracking)
- **August 2025:** Object detection initialization, ML infrastructure setup
- **September 2025:** Progress consolidation, preliminary results compilation

Status: COMPLETED (reporting). Monthly reporting was consistent, though depth of progress documentation varies across months.

3.3 Phase 3: Integration and Advanced Development (September – October 2025)

3.3.1 Quarterly Progress Review (September 4, 2025)

A comprehensive quarterly presentation was delivered, likely consolidating work from Q2 and Q3.

Status: COMPLETED (checkpoint). This represents a formal progress review and planning milestone.

3.3.2 Fellowship and Advanced Architectures (September 18, 2025)

Work on fellowship applications and exploration of advanced neural architectures (MAE, Vision Transformers).

Status: IN PROGRESS. This represents future-oriented work, with preliminary MAE results showing promise.

3.3.3 Existing Radar Architectures Survey (October 16, 2025)

A survey of existing commercial and research radar architectures was conducted.

Status: COMPLETED. This provides valuable context for system design choices.

3.3.4 Annotation Pipeline Development (October 23, 2025)

Planning and preliminary work on the radar-assisted image labeling pipeline.

Status: PARTIALLY COMPLETED. Concepts defined; implementation appears in early stages.

3.3.5 Latest Progress Summary (October 14, 2025)

A summary presentation consolidating August-September progress.

Status: COMPLETED (checkpoint). Recent work synthesized for evaluation.

3.3.6 Radar Software Updates (October 18, 2024 – Note: Likely October 18, 2025)

Work on radar system software and configuration.

Status: PARTIALLY COMPLETED. Appears to involve ongoing system tuning rather than fully formalized results.

3.4 Undergraduate Collaboration and Onboarding (August – October 2025)

UG Presentations (August 26 – October 7, 2025):

Several presentations to undergraduate team members (Cancilla, Kirk) documenting:

- Project kickoff and overview (August 26)
- Iterative progress updates (September 2, 16, 30; October 7)

Status: COMPLETED (team coordination). This indicates leadership in coordinating team members and knowledge transfer.

4 Detailed Technical Accomplishments

4.1 Completed Work

4.1.1 1. Conceptual Framework and System Architecture

Completion: FULLY COMPLETED

The candidate developed a comprehensive system architecture clearly articulating:

- Sensor modalities and their complementary strengths (radar: all-weather range; camera: high-res classification)
- Coordinate transformation requirements (polar radar to Cartesian image coordinates)
- Multi-level fusion approach: direct radar detection, camera verification, joint optimization
- Automated labeling pipeline where radar propositions guide human annotation

This conceptual work is well-documented in the March presentation and refined project proposal document.

4.1.2 2. Comprehensive Literature Review

Completion: FULLY COMPLETED

A 131-reference literature review synthesizing:

- Radar physics and signal processing
- Computer vision for maritime object detection
- Sensor fusion techniques
- Deep learning for radar applications

The review demonstrates mastery of foundational knowledge and identifies key research gaps motivating this work.

4.1.3 3. CFAR Detection Implementation

Completion: SUBSTANTIALLY COMPLETED

Constant False Alarm Rate detector implemented and tested on real Lake Murray radar data:

- Implementation of CA-CFAR algorithm

- Testing on real radar captures showing detection of land boundaries and water surfaces
- Analysis of threshold adaptation across range-azimuth cells

Real data results are presented (Figure 7), validating implementation.

4.1.4 4. Multi-Gain Sampling Methodology

Completion: SUBSTANTIALLY COMPLETED

Development of multi-gain data acquisition strategy:

- Theoretical framework for capturing 10 radar frames at different gain levels
- Demonstration that objects invisible at low gain become visible at high gain (and vice versa)
- Conceptual motivation for feeding multi-channel radar data to CNNs
- Preliminary results showing gain-stacked data enables better target visibility (Figure 11)

The approach is well-motivated and partially validated, though end-to-end CNN integration appears incomplete.

4.1.5 5. Sensor Rig Integration

Completion: SUBSTANTIALLY COMPLETED

- Hardware integration of X-band marine radar (9.4 GHz) with optical camera
- Deployment on Lake Murray for data collection
- Timing synchronization mechanism to pair radar sweeps with camera frames
- Proposed improvement (EZ-Setup system) for future deployments

Current system functional for data collection. Improved version designed but not yet deployed.

4.1.6 6. Data Collection Initiative

Completion: SUBSTANTIALLY COMPLETED

- Pilot dataset collection at Lake Murray (less than 10 hours radar PPI + camera video as target)
- Coordination of collection efforts with team members
- Planning for expansion to multiple SC lakes (7-10 planned locations per Figure 13)
- Strategic approach to capture diverse sea states and lighting conditions

Initial Lake Murray collection appears complete or near-complete. Expansion planning well-developed.

4.2 Partially Completed Work

4.2.1 1. Sensor Calibration and Co-Registration

Completion: IN PROGRESS

Critical for projecting radar detections onto camera images, but calibration is notoriously challenging:

- Extrinsic calibration between radar and camera frames identified as necessary
- Calibration challenges documented (timing errors, different resolutions)
- Autonomous calibration approach planned but implementation status unclear

This remains an open challenge requiring further work.

4.2.2 2. CNN Model Development for Radar Detection

Completion: EARLY STAGE / IN PROGRESS

- Architecture selection discussed (1D convolution, 2D CNN, potential Vision Transformers)
- Comparison against CFAR baseline planned
- No results showing CNN outperforming CFAR presented in available documentation
- MAE self-supervised pre-training explored (preliminary, macro results only)

While foundational work is done, practical CNN performance on radar detection not yet demonstrated.

4.2.3 3. End-to-End Fusion Pipeline

Completion: IN PROGRESS

Integration of radar CNN + camera system:

- Pipeline architecture conceptualized
- User interface design considered
- Integration with camera visualization planned
- Implementation appears to be underway but results not yet published

4.2.4 4. Radar-Assisted Image Labeling Tool

Completion: IN PROGRESS

- Concept fully developed (radar generates region proposals for annotator)
- Planning completed (October 23 presentation on annotation strategy)
- Implementation appears not yet finalized
- Effectiveness metrics planned but not yet collected

4.2.5 5. A-MARV Dataset Development

Completion: IN PROGRESS / PLANNING

- Dataset concept designed (Amplitude-only Maritime Radar Video)
- Collection locations planned across US coast (Figure 16)
- Data format and labeling strategy defined
- Version 0.1 collection locations identified
- Full-scale deployment and release planned for future phases

This is primarily planning; execution will span future work.

4.3 Work in Early Exploration Stage

4.3.1 1. Radar Entropy Analysis

Completion: EXPLORATORY

Entropy explored as potential clutter characterization metric, but integration into detection pipeline not established.

4.3.2 2. Integer and Efficient Representations

Completion: EXPLORATORY

Multiple presentations on integer processing and numerical efficiency, but relationship to main pipeline unclear.

4.3.3 3. Advanced Architectures (Vision Transformers, MAE)

Completion: EARLY STAGE

Preliminary MAE results shown, but not yet integrated into end-to-end system.

5 Work Quality Assessment

5.1 Strengths

5.1.1 Systematic Problem Decomposition

The candidate demonstrated strong ability to break down a complex multi-disciplinary problem into manageable components:

- Radar signal processing (gain, CFAR, clutter modeling)
- Sensor calibration
- Machine learning model development
- Data collection and curation
- Integration and evaluation

Each component was addressed with appropriate technical depth.

5.1.2 Theoretical Foundation

Extensive literature review and conceptual development preceded implementation, indicating strong theoretical grounding. The candidate can articulate why specific approaches (e.g., multi-gain fusion) are technically motivated.

5.1.3 Hands-On Implementation

Rather than purely theoretical work, the candidate built actual systems:

- Functional sensor rig with radar and camera
- Real data collection campaigns

- CFAR detector implementation and validation
- Prototype machine learning frameworks

This is more valuable than simulation-only work.

5.1.4 Iterative Refinement

Multiple presentations on the same topic areas (e.g., gain analysis, integer processing) suggest iterative improvement and exploration rather than one-shot attempts. The trajectory from initial concept to prototype demonstrates learning and adaptation.

5.1.5 Clear Communication

Consistent monthly reports and frequent presentations indicate commitment to documentation and team communication. Presentation materials are generally well-organized and clearly articulate technical concepts.

5.1.6 Cross-Functional Knowledge

The work spans radar signal processing, computer vision, deep learning, and systems integration—demonstrating breadth uncommon in early-stage researchers.

5.2 Areas of Concern / Limitations

5.2.1 Incomplete Integration

Many components are developed in relative isolation. For example:

- CFAR detector results shown independently of CNN development
- Multi-gain data capture methodology not clearly integrated with final detection system
- Calibration challenges identified but solutions not demonstrated

The “end-to-end” system appears to still be under integration.

5.2.2 Limited Quantitative Validation

While many approaches are proposed, there is limited clear evidence of comparative performance. For instance:

- CNN detection results not shown to outperform CFAR baseline
- Multi-gain fusion benefit quantified theoretically but not empirically demonstrated on full detection task

- Labeling efficiency improvements not yet measured

5.2.3 Scope Creep

The project encompasses many advanced topics (Vision Transformers, GPU sharding, Masked Autoencoders, entropy analysis, integer representations). While this demonstrates breadth, the main detection and labeling objectives could be more focused. Not all explorations are clearly contributing to core milestones.

5.2.4 Hardware/Environmental Dependencies

Data collection depends on favorable lake conditions, radar equipment availability, and team coordination. Some work appears delayed or dependent on external factors.

5.2.5 Documentation Density Varies

While monthly reports and presentations are consistent, their depth varies. Some months/presentations provide high-level progress; others lack specific quantitative metrics or failure analysis.

5.2.6 Limited Discussion of Challenges

Specific difficulties (e.g., calibration errors, clutter complexity in rough seas) are identified but recovery strategies not always clearly articulated.

6 Project Timeline and Schedule Performance

The project appears to follow an approximately one-year roadmap (visible in documents) with four quarters:

- **Q1 (Jan-Mar 2025):** Project conception, initial planning
- **Q2 (Apr-Jun 2025):** Literature review, system design, initial data collection
- **Q3 (Jul-Sep 2025):** Technical development, exploration, prototype building
- **Q4 (Oct-Dec 2025):** Integration, evaluation, refinement

Assessment: ROUGHLY ON SCHEDULE

By October 2025:

DONE Literature review and foundational work complete

DONE Sensor rig operational and data collection underway

DONE CFAR detection working on real data

DONE Multi-gain methodology conceptualized and partially validated

TBD CNN models in development but performance not yet proven

TBD Full calibration not completed

TBD End-to-end pipeline not yet fully integrated

The project appears roughly aligned with a 12-month master's thesis timeline, though key integration milestones should be approaching completion in Q4.

7 Characteristics and Traits Relevant to Recommendation

7.1 Positive Traits

7.1.1 1. Technical Depth and Breadth

The candidate demonstrates mastery across multiple technical domains:

- **Radar signal processing:** Understands gain control, CFAR, clutter modeling, Doppler filtering
- **Computer vision:** Familiar with object detection architectures (YOLO, R-CNN)
- **Deep learning:** Exploring CNNs, Vision Transformers, self-supervised approaches (MAE)
- **Systems engineering:** Integrating hardware sensors, implementing calibration, coordinating data collection

This is unusual for a Master's student and indicates strong foundational preparation or self-directed learning.

7.1.2 2. Problem-Driven Research

Rather than pursuing algorithms for their own sake, the candidate articulates specific motivations:

- Radar alone misses small targets in clutter → propose sensor fusion
- Camera-based detection lacks labeled maritime data → propose radar-assisted labeling
- Single gain fails across dynamic conditions → propose multi-gain stacking

Each technical choice is motivated by concrete problem requirements.

7.1.3 3. Willingness to Build

The candidate moved beyond theory to construct actual systems:

- Deployed sensor hardware and coordinated data collection
- Implemented signal processing algorithms on real data
- Set up machine learning infrastructure

This practical experience is valuable and difficult to acquire.

7.1.4 4. Systematic Planning and Organization

The work follows clear structure:

- Conceptual design before implementation
- Incremental milestones and deliverables
- Regular progress reporting and team communication
- Planned dataset expansion and future directions

7.1.5 5. Strong Communication Skills

Frequent presentations and reports demonstrate ability to articulate technical work to diverse audiences. Diagrams are clear; explanations are systematic.

7.1.6 6. Adaptability and Iterative Learning

Multiple presentations on the same topics (gain analysis, integer processing) and exploration of diverse techniques (CFAR, CNN, MAE) suggest willingness to refine approaches and learn from iteration.

7.1.7 7. Vision for Future Work

The candidate articulated ambitious but plausible next steps (self-supervised learning, A-MARV dataset, physics-informed models, continual learning). This indicates strategic thinking beyond immediate task completion.

7.2 Areas for Development

7.2.1 1. Completion and Integration Focus

While many components are well-developed, integration into a unified end-to-end system is incomplete. Recommendation: Prioritize completing the detection pipeline and demonstrating superior performance over baselines, even if not all advanced features are included.

7.2.2 2. Quantitative Rigor

Claims about performance improvements could be more rigorously validated. Recommendation: Generate clear performance metrics comparing proposed approaches to baselines (e.g., CNN vs. CFAR detection rates, labeling efficiency gains).

7.2.3 3. Scope Management

The breadth of exploration is commendable but could benefit from tighter focus. Recommendation: Identify the 3-4 most critical technical challenges and allocate resources accordingly. Some explorations (e.g., GPU sharding, entropy analysis) may be premature.

7.2.4 4. Failure Analysis and Problem-Solving

While challenges are identified (e.g., calibration difficulty, clutter in rough seas), responses are sometimes vague. Recommendation: When obstacles are encountered, document specific diagnostic steps and attempted solutions. This demonstrates problem-solving skills.

7.2.5 5. Literature Connection

The literature review is comprehensive, but subsequent work could more explicitly position results relative to prior art. Recommendation: Systematically compare CNN performance to published radar detection algorithms.

8 Trends and Observations

8.1 Technical Trajectory

The work shows clear progression:

1. **Foundation** (Mar-May): Conceptualization and literature
2. **Exploration** (Jun-Aug): Multiple technical approaches investigated in parallel

3. **Integration** (Sep-Oct): Consolidation and preliminary system assembly
4. **Validation** (expected Q4): Performance evaluation and refinement

This is a healthy research trajectory suggesting learning and maturation.

8.2 Increasing Complexity

Early work focuses on foundational topics (gain control, CFAR, basic sensor calibration). Later work addresses advanced techniques (Vision Transformers, self-supervised learning, continual learning). This increasing sophistication is appropriate.

8.3 Scope Evolution

Initial scope was “sensor fusion for detection + automated labeling.” Scope expanded to include:

- Multi-gain radar processing
- Novel applications of Masked Autoencoders
- A-MARV dataset development
- GPU-optimized implementations

This indicates ambition and confidence, though focus would be beneficial.

8.4 Team Leadership

Undergraduate students (Cancilla, Kirk) are being onboarded and trained. This suggests emerging leadership and ability to communicate complex ideas to less-experienced researchers.

9 Completion Status Summary

9.1 Overall Progress

Approximately 50-60% of the envisioned one-year project appears complete. Foundational and architectural work is solid. Implementation and integration of the core system is underway but not finalized. Advanced features are exploratory.

For a Master’s thesis timeline (typically 12-18 months), progress as of October 2025 is reasonable, though completion of key integration milestones should be prioritized in the final quarter.

Component	Status	Notes
Project Concept	[DONE]	Well-articulated, clearly motivated
Literature Review	[DONE]	131+ references, comprehensive
System Architecture	[DONE]	Detailed design with clear dataflow
Sensor Hardware Integration	[DONE]	Functional radar + camera rig
Data Collection (Pilot)	[DONE]	Lake Murray dataset acquired
CFAR Detection	[DONE]	Implemented, validated on real data
Multi-Gain Methodology	[PARTIAL]	Conceptualized, partially validated
CNN Model Development	[PARTIAL]	Early stages, no baseline comparison
Sensor Calibration	[PARTIAL]	Challenges identified, solutions not finalized
End-to-End Pipeline	[PARTIAL]	Architecture designed, integration underway
Labeling Tool	[PARTIAL]	Concepts designed, implementation in progress
A-MARV Dataset	[PARTIAL]	Planning complete, v0.1 not yet deployed

10 Recommendations for Reference Letter

10.1 Key Strengths to Emphasize

1. **Technical Versatility:** The candidate demonstrates competence across radar signal processing, computer vision, deep learning, and systems engineering. This is rare and valuable.
2. **Problem-Solving Orientation:** Work is motivated by concrete challenges. The candidate doesn't pursue interesting techniques in isolation; rather, each technical choice addresses a specific need.
3. **From Theory to Practice:** The candidate moved beyond simulation to build real systems, collect actual data, and validate on hardware. This practical experience is invaluable.
4. **Research Vision:** The articulated future directions (self-supervised learning, A-MARV dataset, physics-informed models) indicate strategic thinking and ability to identify impactful research questions.
5. **Communication and Leadership:** Regular presentations, clear documentation, and team onboarding demonstrate strong communication and emerging leadership.

10.2 Contextual Caveats

1. Much work remains in-progress. The integration of core components (CNN detection + sensor fusion + labeling) is not yet complete. Recommend noting this is typical for research at this stage.
2. Quantitative validation is incomplete. While approaches are promising, performance advantages are not yet rigorously demonstrated. This is an area for future

emphasis.

3. The project scope is ambitious. Successful completion of the core system (detection + labeling) would be a significant achievement even if all planned advanced features are not finalized.

10.3 Suggested Framing

A strong recommendation could emphasize:

“The candidate has demonstrated exceptional breadth and depth across signal processing, computer vision, and systems engineering. Rather than pursuing well-trodden paths, they identified a novel gap at the intersection of radar and optical sensing for maritime applications, and designed a creative solution combining sensor fusion with automated data labeling. While the research is still in progress, the foundational work is solid, the technical approach is well-motivated, and the candidate has shown willingness to tackle both theoretical challenges and practical implementation. Their combination of technical skill, problem-driven thinking, and communication ability makes them well-suited for continued graduate study or industry research roles.”

11 Conclusion

Over approximately seven months (March–October 2025), the candidate has conducted systematic research on radar-camera fusion for maritime object detection. The work spans conceptualization, literature review, system design, prototype development, data collection, and initial model training.

Key accomplishments include:

- Comprehensive literature review establishing research foundation
- Well-articulated system architecture addressing fundamental sensor fusion challenges
- Functional sensor rig and initial data collection pipeline
- Implementation of classical (CFAR) and early machine learning detection approaches
- Planning for large-scale dataset (A-MARV) to support future research

Current status: Approximately 50-60% of planned first-year work is complete. Foundational components are strong. Integration and validation of the core system (detection + automated labeling) is underway.

Strengths: Technical depth, problem-driven design, willingness to build and test, clear communication, strategic vision.

Areas for continued focus: Completing end-to-end system integration, quantitative validation of performance claims, prioritized execution of remaining milestones.

The candidate exhibits strong research potential and would be well-positioned for continued graduate study, industry research roles, or both. The combination of technical skill, systems thinking, and problem-driven innovation is notable.

A Document References

This report synthesizes information from the following source documents:

Project Presentations:

- 03_02_25_INITIAL_PROJECT_IDEA.pdf
- 06_24_25_Initial_team_weekly_meeting.pdf
- 06_26_25_gain_timing.pdf
- 07_02_25_integer.pdf
- 07_07_25_HDR_gain_comparison.pdf
- 07_21_25_radar_entropy.pdf
- 07_30_25_Radar_calibration-Radar_simulation.pdf
- 08_14_25_Object_Detection.pdf
- 08_19_25_object_tracking-MAE-ML_stuff.pdf
- 09_04_25_QPR_presentation.pdf
- 09_18_25_Fellowships_MAE_architecture_sliding.pdf
- 09_23_25_GPU_sharding.pdf
- 09_25_25_GPU_sharding_libraries.pdf
- 10_09_25_weekly.pdf
- 10_14_25_summary_sept-aug_progress.pdf
- 10_16_25_existing_radar_architectures.pdf
- 10_18_24_Radar_software_updates.pdf

- 10_22_25_Integer_pres.pdf
- 10_23_25_plan_for_annotation.pdf

Progress Reports:

- Phase2_May2025_VAUGHT.pdf
- Phase2_June2025_VAUGHT.pdf
- Phase2_July2025_VAUGHT.pdf
- Phase2_August2025_VAUGHT.pdf
- Phase2_September2025_VAUGHT.pdf
- annotation_LR_10_18_25.pdf

Foundation Documents:

- Lit_review_may2025.pdf (comprehensive literature review)

Team Coordination:

- UG_presentations folder (team onboarding documents)